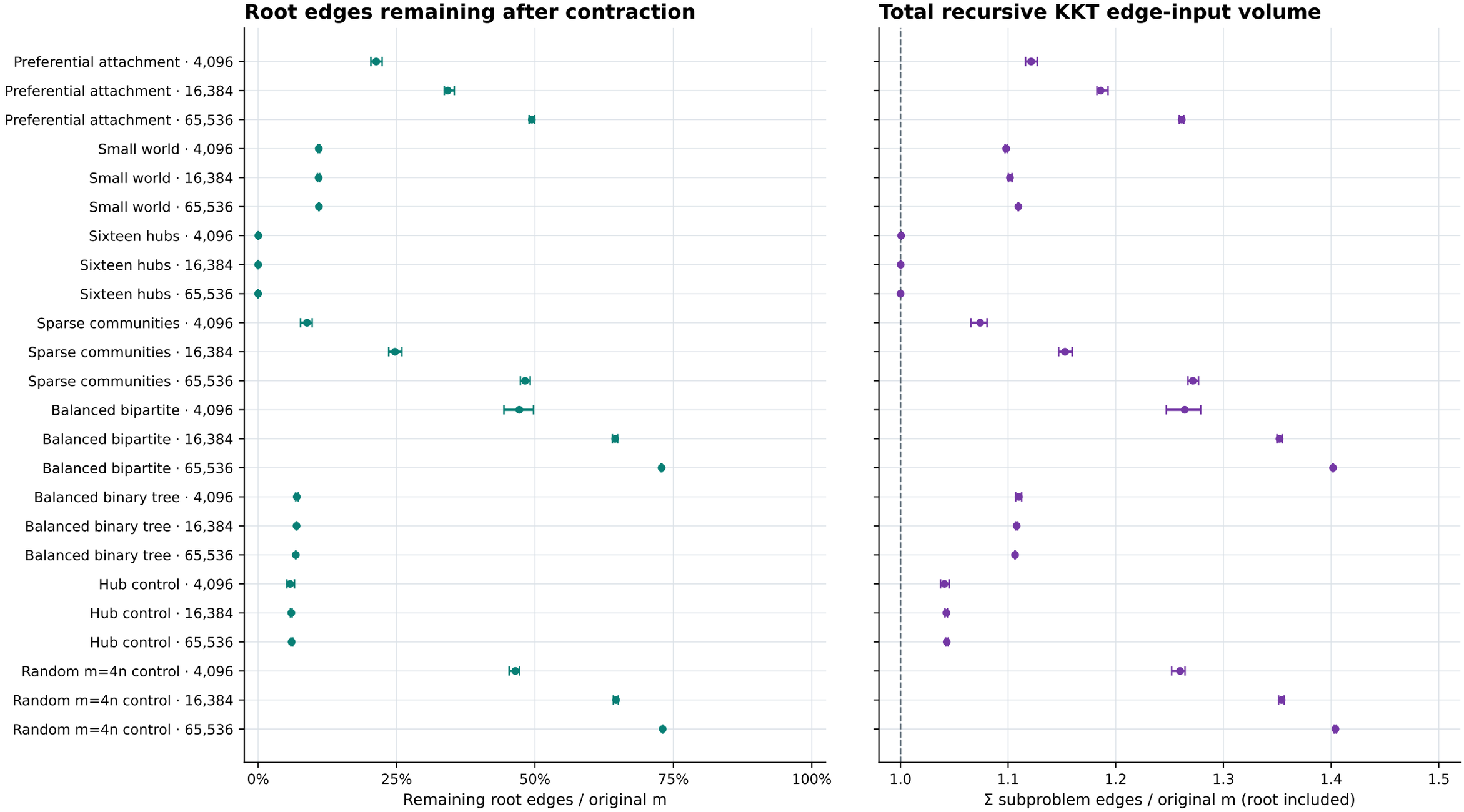
# Separate untimed diagnosis of contraction and recursion




Arithmetic seed means; bars span graph-level means. Zero remaining edges includes an earlier one-round finish.
Edge-input volume excludes repeated internal passes. These counters do not isolate a causal sampling speedup.